```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Asus> javac Assignment.java
PS C:\Users\Asus> java Assignment
Logic 1:
     X
    X X
   X   X
  X     X
 X       X
X         X
 X       X
  X     X
   X   X
    X X
     X

Logic 2:
     X
    X X
   X   X
  X     X
 X       X
X         X
 X       X
  X     X
   X   X
    X X
     X

Logic 3:
     X
    X X
   X   X
  X     X
 X       X
X         X
 X       X
  X     X
   X   X
    X X
     X


Logic 1:
Size must be odd.

Logic 2:
Size must be odd.

Logic 3:
Size must be odd.
PS C:\Users\Asus> |
```

```
Windows PowerShell
|  Welcome to JShell -- Version 25.0.1
|  For an introduction type: /help intro

jshell> int a = 'A'
a ==> 65

jshell> int a = 'A';a+4
a ==> 65
$3 ==> 69

jshell> a= 5
a ==> 5

jshell> -a-++a+--a
$5 ==> -6

jshell> String b = "A\tb c\sd"
b ==> "A\tb c d"

jshell> b.replace(" ","")
$7 ==> "A\tbcd"

jshell> Byte.SIZE+Short.SIZE/Short.Bytes
|  Error:
|  cannot find symbol
|    symbol:   variable Bytes
|  Byte.SIZE+Short.SIZE/Short.Bytes
|                       ^--------^

jshell> short(5022026)
|  Error:
|  ';.class' expected
|  short(5022026)
|        ^
|  Error:
|  unexpected type
|    required: value
|    found:    class
|  short(5022026)
|  ^---^

jshell> Short(5022026)
|  Error:
|  cannot find symbol
|    symbol:   method Short(int)
|  Short(5022026)
|  ^---^

jshell> (Short)5022026
|  Error:
|  incompatible types: int cannot be converted to java.lang.Short
|  (Short)5022026
|         ^-----^

jshell> (short)5022026
$8 ==> -24246
```

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Practical No. -

Q.1

a) int a = 'A' ; a+4

a = 65 coz ASCII value of 'A' is 65.

a + 4 ⇒ 65 + 4 ⇒ 69

b) a = 5

-a - ++a + --a

⇒ (-a) - (++a) + (--a)

[++a = a+1 = 5+1 = 6
--a = a-1 = 6-1 = 5]

⇒ (-5) - (6) + (-5)

⇒ -6 Ans

c) String b = "A\tb c\sd"

b.replace (" ", "")

replace method replaces a particular character of string with the given character.

∴ " " will be replaced by ""

⇒ A\tb c\sd

d) Byte.SIZE + short.SIZE / short.Bytes

↳ ERROR.

coz Bytes is used instead of BYTES. it should be all in capital. java is case sensitive.

Q.2 a)

```
public class Pattern {
    public static void printPattern (int size) {
    if (size % 2 == 0) {
    System.out.println ("The entered value"
    + size + " is not odd. Please enter an
        odd size");
    return;
    }

    for (int i = 0; i < size; i++) {
        for (int j = 0; j < size; j++) {
        if (j == i || j == size - 1 - i ) {
        System.out.print("X");
        } else {
        System.out.print(" ");
        }
    }
    System.out.println();
    }
}

public static void main (String[] args)
{
    printPattern(5);
}
}
```

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Q.2 b) (Short) 5022026

short's range is -2^{15} to $2^{15}-1$ i.e.

-32768 to 32767

5022026 exceeds this range, so there is an overflow of data. To bring it in the range we gotta take it's modulo w.r.t 2^{16} ∵ short is of 16 bytes.

$5022026 \% 2^{16} = \cancel{368970}.41290$

(If multiplied the quotient is 76)

~~5022~~ Whenever there is an overflow of data it returns to the ~~to~~ smallest no. and starts from bottom.

32767
32766
⋮
-3
-2
-1
0
1
2
3
⋮
-32767
-32768

~~Answer~~ ∵ ~~41290 − 32767 = 8523~~

∵ ~~−32768 + 8523 = −24,245~~

∵ $41290 - \cancel{327}\,2^{16} = -24246$

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Q.2 c)

isBlank() method of the String class can't always be used as a replacement for isEmpty(). isBlank doesn't detect white spaces while isEmpty does.

	isBlank()	isEmpty()
""	True	True
" _ " (1 space)	True	False
" _ _ " (2 space)	True	False.
"abc"	False	False.